



PROJECT REPORT

Cricket Player Performance Tracker

A Salesforce.com Application for Recording, Monitoring and Analyzing Player
Performance

Platform	Salesforce CRM (Lightning Experience)		
Project Type	Cloud-based sports performance tracking system		
Prepared for	Academic Project / Practical Submission		
Team members. Team Leader.	Name	Role	NM ID
	ARUN R	Member	astvu32332324u09004
	S KISHORE	Team Lead	astvu32332324u09073
	HARIHARAN S	Member	astvu32332324u09022
	CHANDRU S	Member	astvu32332324u09015
	KARTHIK K	Member	astvu32332324u09031
Department / College	VOORHEES COLLEGE / BCA		
Academic Year	2026–2027		

Project scope: This report describes a practical Salesforce configuration that can be built using custom objects, fields, relationships, validation rules, reports, dashboards and automation. It is a proposed implementation blueprint; actual deployment and test results should be recorded after building it in a Salesforce org.

Abstract

The Cricket Player Performance Tracker is a Salesforce-based application intended to maintain organized records of cricket players, matches and individual match performances. It helps coaches, team managers and authorized staff enter performance statistics, review player progress, and view summary reports from a centralized CRM interface. The proposed solution uses Salesforce custom objects and relationships, validation rules to improve data quality, reports and dashboards for analysis, and optional Flow automation for routine tasks.

The system focuses on dependable record keeping and clear presentation of statistics. It does not automatically judge player ability; performance interpretation remains with coaches and authorized users.

Keywords: Salesforce, CRM, Cricket, Player Statistics, Custom Objects, Reports, Dashboard, Flow.

Table of Contents

1. Introduction
 2. Problem Statement and Objectives
 3. Scope and Users
 4. System Requirements
 5. Salesforce Data Model
 6. Application Workflow
 7. Implementation Steps
 8. Reports and Dashboard
 9. Testing Plan
 10. Security and Data Quality
 11. Benefits and Limitations
 12. Future Enhancements
 13. Conclusion
- Appendix A: Sample Fields and Formulas

1. Introduction

Cricket teams generate match-by-match information such as runs, balls faced, wickets, overs, runs conceded, catches and match results. When these details are stored in separate spreadsheets or paper records, staff may find it difficult to maintain consistent player histories or produce quick summaries. Salesforce provides a configurable cloud platform where related records can be stored, searched and reported through a common interface.

This project proposes a Salesforce application to capture player profiles and their performances in each match. Users can open a player record, review related match-performance entries, and use reports or dashboards to examine trends over a selected period or competition.

2. Problem Statement and Objectives

Problem statement: Cricket player statistics may be scattered across manual records, making it harder to maintain accurate histories, find match details and prepare consistent performance summaries.

Objectives:

- Create a centralized player register.
- Record match details and each player's performance for that match.
- Reduce incomplete or invalid entries using required fields and validation rules.
- Provide searchable lists, reports and dashboards for coaches and team staff.
- Support secure access based on user roles and permissions.

3. Scope and Users

The initial version is designed for one team or academy, with the option to extend to multiple teams and competitions. It records data entered by authorized users; live scoring, ball-by-ball tracking, external statistics feeds and predictive player ratings are outside the initial scope.

Primary users

- Administrator: manages configuration, users, access and data quality.
- Coach / Team Manager: reviews player histories, match summaries and dashboards.
- Scorer / Data Entry User: creates match and performance records.
- Read-only Viewer: views approved records and reports.

4. System Requirements

Platform: Salesforce org with Lightning Experience and permission to create custom objects, fields, reports, dashboards and (if used) Flows.

Client: Current supported desktop or mobile browser; Salesforce mobile app may be used if layouts are configured.

Functional: Create and update player, match and performance records; link performance to a player and match; search and filter; run reports; view dashboard summaries.

Non-functional: Role-based access, consistent data, understandable page layouts, maintainable configuration and regular data backup/export according to organizational policy.

5. Salesforce Data Model

Create the following custom objects. The suggested API names are examples; use the actual names generated in the Salesforce org consistently.

Object	Suggested API name	Purpose / Key fields
Player	Player__c	Player Name, Player ID, Date of Birth (optional), Batting Style, Bowling Style, Primary Role, Team, Active Status
Match	Match__c	Match Name, Match Date, Opponent, Competition, Venue, Format, Result
Player Performance	Player_Performance__c	Player (Lookup/Master-Detail), Match (Lookup/Master-Detail), Runs, Balls Faced, Fours, Sixes, Overs, Maidens, Runs Conceded, Wickets, Catches, Run Outs
Team (optional)	Team__c	Team Name, Coach, Season; associate players and matches when multi-team support is needed

Relationships: One Player can have many Player Performance records. One Match can have many Player Performance records. Each performance record links one player to one match. A unique external ID or duplicate rule can be considered to reduce duplicate player-match entries.

6. Application Workflow

1. Administrator configures objects, fields, page layouts, permissions and picklist values.
2. Data-entry user creates or updates the Player record.
3. User creates a Match record with date, opponent, format and venue.
4. After the match, user creates one Player Performance record per participating player and links it to the correct Player and Match.
5. Validation rules check basic data consistency before saving.
6. Coach opens the player record or runs reports to review performance over time.
7. Dashboard charts summarize runs, wickets and participation for the selected reporting period.

7. Implementation Steps

Step 1 — Create custom objects: In Setup, open Object Manager and create Player, Match and Player Performance objects. Enable reports and activities only where useful.

Step 2 — Add fields: Create the fields listed in Appendix A. Use Number fields for counts and Decimal fields for overs if overs are stored in cricket notation. Prefer storing balls bowled as a whole number if accurate calculations are required.

Step 3 — Configure relationships: Add Player and Match relationship fields to Player Performance. Choose Master-Detail only if the organization wants parent-controlled ownership/sharing and cascade delete; otherwise use Lookup relationships.

Step 4 — Build Lightning pages: Add key fields, related lists and compact layouts. Place the Player Performance related list on Player and Match pages.

Step 5 — Add validation: Prevent negative statistics, require essential links, and ensure balls faced, fours and sixes are plausible. Avoid overly strict rules that reject legitimate scoring situations.

Step 6 — Configure reports and dashboard: Create custom report types if needed, then build summary reports and dashboard components.

Step 7 — Test with sample records: Enter a few fictional players, matches and performance rows, verify calculations and access, and correct page layouts before wider use.

8. Reports and Dashboard

Suggested reports:

- Player Match-by-Match Performance — player, match date, opponent, runs, balls faced, wickets and catches.
- Top Run Scorers — grouped by player with sum of runs for a selected period.
- Wicket Summary — total wickets by player and competition.
- Match Participation — count of performance records by match.
- Player Performance by Format — compare records across T20, ODI, Test or other configured formats.

Suggested dashboard components:

- Total runs by player (bar chart).
- Total wickets by player (bar chart).
- Runs by match date (line chart).
- Matches recorded and active player count (metric components).
- Filters for season, competition, format and player where supported.

9. Testing Plan

Test case	Action	Expected result
Create player	Save a player with required fields.	Record saves and appears in player list.
Link performance	Create performance linked to a player and match.	Record appears in both relevant related lists.
Negative statistic	Enter -5 runs or -1 wicket.	Validation blocks invalid values.
Missing relationship	Attempt to save performance without required player or match.	Required-field validation prevents save.
Report totals	Create known sample performances and run totals report.	Displayed sums match entered test data.
Access control	Log in as read-only user and attempt record edit.	User can view permitted data but cannot edit.

Record actual outcomes, tester name, date and any defects in the final submission after executing these tests in your Salesforce org.

10. Security and Data Quality

Use profiles and permission sets to grant only the object and field access each role needs. Restrict deletion of match and performance records to trusted administrators. Review organization-wide defaults, sharing rules and field-level security. Avoid storing unnecessary sensitive personal data; date of birth should be optional and access-limited if collected. Use consistent picklist values for match format, role and result. Establish a process for correcting errors and periodically exporting or backing up records according to institutional policy.

11. Benefits and Limitations

Expected benefits: Centralized records, quicker retrieval, repeatable reporting, clearer player histories, and less manual consolidation of statistics.

Limitations: Data quality depends on accurate entry; Salesforce licensing and org permissions may constrain available features; advanced scoring and analytics may require additional configuration or integrations; the proposed solution does not replace official scorekeeping rules or coaching judgment.

12. Future Enhancements



Possible extensions include a mobile-friendly scoring form, season and tournament management, automated reminders for incomplete match entries, CSV import templates, integration with an authorized scoring data source, and trend charts using carefully defined metrics. Any player comparison should use consistent match formats and sufficient context.

Field label	Type / Formula	Suggested rule / note
Runs	Number (0 decimals)	Minimum 0; maximum appropriate to competition policy.
Balls Faced	Number (0 decimals)	Minimum 0.
Fours / Sixes	Number (0 decimals)	Minimum 0; validate boundaries according to match rules.
Wickets	Number (0 decimals)	Minimum 0; use a competition-appropriate maximum.
Overs Bowled	Number (2 decimals)	Cricket over notation is not ordinary decimal notation; prefer total balls for calculations.
Balls Bowled	Number (0 decimals)	Recommended for reliable economy-rate calculations.
Runs Conceded	Number (0 decimals)	Minimum 0.
Batting Strike Rate	IF(Balls_Faced__c > 0, Runs__c * 100 / Balls_Faced__c, 0)	Formula result: Number, 2 decimals.
Bowling Economy	IF(Balls_Bowled__c > 0, Runs_Conceded__c * 6 / Balls_Bowled__c, 0)	Formula result: Number, 2 decimals; assumes total legal balls bowled.

The Cricket Player Performance Tracker demonstrates how Salesforce custom objects, relationships, validation rules, reports and dashboards can support structured cricket statistics management. By linking each performance to a player and match, the system creates a useful record history and supports summary reporting. The application can be expanded as team requirements grow, while access controls and data validation help maintain reliable records.

Sample Fields and Formulas

**IF(Matches_Played__c > 0,
Total_Runs__c / Matches_Played__c ,
0)**